

COPD — Air Traffic Control Tower (Diagnosis, Gas Exchange & GOLD Staging)



1. V/Q mismatch radar

WHAT YOU SEE

Green radar screen banner reading 'V/Q mismatch = #1 cause of low PaO2'.

WHAT IT ENCODES

Ventilation-perfusion (V/Q) mismatch is the PRIMARY mechanism of hypoxemia in COPD: destroyed alveolar units and uneven airflow create regions of low V/Q (perfusion without ventilation) and high V/Q (dead space). This widens the A-a gradient. As disease advances, alveolar hypoventilation and reduced DLCO (in emphysema) contribute, but mismatch dominates early hypoxemia.

BOARD TRAP

The main cause of low PaO2 in COPD is V/Q mismatch — NOT pure hypoventilation, a diffusion defect, or an anatomic shunt. A normal A-a gradient (pure hypoventilation) is the wrong attribution.

2. Responds to O2 / shunt crossed out

WHAT YOU SEE

Two crossed-out signs ('SHUNT', 'DIFF') with notes that V/Q mismatch responds to supplemental O2.

WHAT IT ENCODES

Hypoxemia from V/Q mismatch CORRECTS with supplemental oxygen because raising inspired O2 overcomes the low-V/Q units. This is the key physiologic distinction from a true right-to-left shunt, where deoxygenated blood bypasses ventilated alveoli entirely.

BOARD TRAP

A true right-to-left SHUNT does NOT correct with 100% O2; COPD hypoxemia does respond. If supplemental O2 fails to raise PaO2, think shunt physiology (e.g., pneumonia/ARDS, intracardiac shunt), not simple COPD.

3. FEV1/FVC < 0.7 monitor

WHAT YOU SEE

Central tower monitor showing the obstructive flow-volume curve and 'FEV1/FVC < 0.7'.

WHAT IT ENCODES

The diagnosis of COPD REQUIRES a POST-bronchodilator FEV1/FVC < 0.70 on spirometry — this fixed ratio defines persistent airflow obstruction. The flow-volume loop shows a scooped/concave expiratory limb. FEV1 % predicted is then used to grade severity (GOLD 1-4), not to make the diagnosis.

BOARD TRAP

Diagnosis hinges on the post-bronchodilator FEV1/FVC RATIO, not FEV1 alone, not symptoms, and not the pre-bronchodilator value. Using a fixed 0.7 cutoff can over-diagnose the elderly and under-diagnose the young (where lower-limit-of-normal is more accurate) — a nuance occasionally tested.

4. No entry without spirometry

WHAT YOU SEE

Red 'NO ENTRY without spirometry' barrier sign by the controller.

WHAT IT ENCODES

Spirometry is MANDATORY to confirm COPD; clinical suspicion, symptoms, and imaging cannot substitute. Spirometry also quantifies severity and tracks decline over time. Supporting findings include increased TLC and RV (air trapping/hyperinflation) and reduced DLCO when emphysema predominates.

BOARD TRAP

Do NOT diagnose COPD on chest X-ray, CT, or symptoms alone — a hyperinflated CXR or barrel chest without confirmatory spirometry is insufficient for diagnosis.

5. Irreversible / does not normalize

WHAT YOU SEE

Sign with crossed-out reversibility chart and inhaler, 'DOES NOT normalize / NOT ASTHMA'.

WHAT IT ENCODES

COPD airflow obstruction is largely FIXED — after bronchodilator the FEV1/FVC does not fully normalize (stays <0.7). Some bronchodilator response can occur, but it does not return the ratio to normal. This distinguishes COPD from asthma, where obstruction is typically reversible.

BOARD TRAP

Significant reversibility that NORMALIZES the FEV1/FVC (or >12% and >200 mL FEV1 improvement with full normalization) points to ASTHMA, not COPD. Note: a positive bronchodilator response does not by itself exclude COPD (asthma-COPD overlap exists).

6. PaO2 drops gauge

WHAT YOU SEE

Green-to-red gauge labeled PaO2 with 'NOW drops only' and severity values.

WHAT IT ENCODES

Hypoxemia (falling PaO2) develops and worsens as COPD progresses from V/Q mismatch. Resting hypoxemia is the trigger for evaluating long-term oxygen therapy. Sustained hypoxemia drives pulmonary vasoconstriction and secondary erythrocytosis.

7. PaCO2 rises gauge

WHAT YOU SEE

Second gauge labeled PaCO2 noting it 'RISES' in severe (FEV1 <25-35%) disease.

WHAT IT ENCODES

Hypercapnia is a LATE finding, typically appearing once FEV1 falls below ~25-35% predicted, when ventilatory capacity can no longer eliminate CO2. Chronic CO2 retention is buffered by renal bicarbonate retention, keeping pH near normal.

BOARD TRAP

Early COPD usually has low or normal PaCO2 (from increased ventilation); a RISING PaCO2 signals advanced disease. Acute CO2 rise with acidosis (pH <7.35) signals an exacerbation needing NIPPV, not chronic compensated retention.

8. Cor pulmonale alarm

WHAT YOU SEE

Red flashing 'COR PULMONALE' alarm light near the gas gauges.

WHAT IT ENCODES

Chronic alveolar hypoxia causes pulmonary arteriolar vasoconstriction (and remodeling) → pulmonary hypertension → pressure overload of the RIGHT ventricle = cor pulmonale, manifesting as JVD, hepatomegaly, and peripheral edema. Long-term oxygen therapy can blunt this progression.

BOARD TRAP

Cor pulmonale is RIGHT-heart failure caused by lung disease — not left ventricular failure. Edema in advanced COPD is right-sided in origin; don't attribute it to LV dysfunction by default.

9. ABG with renal compensation

WHAT YOU SEE

Sign showing pH 7.32 / PCO₂ 65 / elevated HCO₃ with 'chronic / renal compensation'.

WHAT IT ENCODES

Chronic respiratory acidosis is buffered by renal RETENTION of bicarbonate, raising HCO₃ to keep pH near normal. The compensation is metabolic and takes 3-5 days to fully develop. A pH of ~7.32 with PCO₂ 65 and high HCO₃ reflects partial compensation of chronic hypercapnia.

BOARD TRAP

Elevated HCO₃ with high PCO₂ and near-normal pH = CHRONIC compensated respiratory acidosis, not an acute respiratory acidosis (which would show a sharply low pH with normal HCO₃). Misreading this as a primary metabolic alkalosis is a classic trap.

10. GOLD 1 — Mild (FEV1 ≥80%)

WHAT YOU SEE

Green runway '1 GOLD' labeled 'FEV1 ≥80% MILD'.

WHAT IT ENCODES

GOLD 1 (mild): post-bronchodilator FEV1 ≥ 80% predicted, in a patient who already meets FEV1/FVC <0.70. Severity grading by FEV1 informs prognosis and monitoring but, importantly, does NOT by itself select initial pharmacotherapy.

BOARD TRAP

Remember the ratio criterion still applies — FEV1 ≥80% only counts as COPD if FEV1/FVC is <0.7. GOLD grade alone doesn't dictate the inhaler regimen (the ABE groups do).

11. GOLD 2 — Moderate (50-79%)

WHAT YOU SEE

Yellow runway '2 GOLD' labeled 'FEV1 50-79%'.

WHAT IT ENCODES

GOLD 2 (moderate): post-bronchodilator FEV1 50-79% predicted. This is often when patients first present with exertional dyspnea and seek care.

12. GOLD 3 — Severe (30-49%)

WHAT YOU SEE

Orange runway '3 SEVERE' labeled 'FEV1 30-49%'.

WHAT IT ENCODES

GOLD 3 (severe): post-bronchodilator FEV1 30-49% predicted. Hypoxemia and exacerbations become more frequent in this range.

13. GOLD 4 — Very severe (<30%)

WHAT YOU SEE

Red runway '4 VERY SEVERE' labeled 'FEV1 <30%'.

WHAT IT ENCODES

GOLD 4 (very severe): post-bronchodilator FEV1 < 30% predicted. This is where chronic hypercapnia, cor pulmonale, and the need for LTOT and advanced therapies (LVRS, transplant) most commonly arise.

14. BODE index trophy

WHAT YOU SEE

Trophy with 'BODE' list: BMI, Obstruction, Dyspnea, Exercise; 'better predictor than FEV1 alone'.

WHAT IT ENCODES

The BODE index combines BMI, airflow Obstruction (FEV1), Dyspnea (mMRC scale), and Exercise capacity (6-minute walk distance) into a multidimensional score that predicts mortality better than FEV1 alone. A low BMI, worse dyspnea, and shorter walk distance each worsen prognosis.

BOARD TRAP

FEV1 alone is an INFERIOR mortality predictor compared with the composite BODE index — picking 'FEV1 best predicts survival' is the trap.